

A hard-prime ES shell missed by first-degree moments: exact counterexample, positive repairs and a progression

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14 September 2026 — research preprint

Abstract

An occupied original divisor shell at a prime congruent to one modulo 840 has negative-definite aggregate first-degree signed localizing form. We give its complete finite residue calculation, deterministic primality certificate, and unique marked witness. Retaining the exterior channel separately or applying a discrete quartic both certify positivity. The witness extends to an explicit all-integer arithmetic progression with infinitely many prime members. This is not an ES counterexample, new prime coverage, or a claim of historical priority.

Original definitions. For $p/4 < a < p/2$ and $R = 4a - p$, each divisor $u \mid a^2$ has two channel labels, with $g_E = 4u + 1$, $g_M = u + a$ and $D = R/\gcd(R, g_c)$. The original rational triples are

$$E : \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad M : \left(a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right).$$

They sum reciprocally to $4/p$ since $(RY - pa)(RZ - pa) = p^2a^2$. The two nonfirst reduced denominators are both D , using $4a \equiv p \pmod{R}$ and $\gcd(pa, R) = 1$. Thus $D = 1$ is exactly a hit, and all failed D are odd and at least three. These are the inherited Type I/II coordinates [1].

Let $\mu = \sum_{c,u} \delta_{1/D}$ retain every original state. Define

$$L_1(\mu) = \max_{\deg P \leq 1, P(1)=1} \frac{3}{2} \int (x - 1/3)P(x)^2 d\mu(x). \quad (1)$$

Successful atoms contribute one and failed atoms nonpositively, so $L_1 \leq H$, where H is the original hit count. The preceding finite screen found no occupied shell with $L_1 \leq 0$ through prime parameter 50000; its source explicitly made no universal converse claim [2].

Theorem 1 (An occupied hard shell with no aggregate first-degree certificate). *Let*

$$p = 19748674681 \equiv 1 \pmod{840}, \quad a = 4937168704 = 2^6 \cdot 13^4 \cdot 37 \cdot 73, \quad R = 135. \quad (2)$$

Then p is prime, the shell contains exactly one original hit, but

$$L_1(E + M) = -\frac{50887012}{144801015} < 0. \quad (3)$$

Proof. A full order certificate uses base 11 and $p - 1 = 2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 23510327$. Every factor is prime by trial division, the largest requiring divisors only through 4848. Direct modular powering gives $11^{p-1} \equiv 1$ and the residues

$\frac{q}{11^{(p-1)/q} \bmod p}$	2	3	5	7	23510327
	19748674680	10039250736	3428156451	13907516733	12674160518.

The gcd with p of every displayed residue minus one is one. Thus modulo any prime divisor of p the order of 11 is divisible by $p - 1$, forcing that divisor to be at least p . This proves primality. The verifier also independently tries every odd divisor of p through 140529.

In the finite group algebra of $(\mathbb{Z}/135\mathbb{Z})^\times$, form

$$\left(\sum_{i=0}^{12} [2]^i \right) \left(\sum_{j=0}^8 [13]^j \right) \left(\sum_{k=0}^2 [37]^k \right) \left(\sum_{l=0}^2 [73]^l \right). \quad (4)$$

Applying the two gates to every residue with its coefficient gives the following complete table. The factors sum to 1053 actual divisors; direct integer divisor enumeration gives the same table independently.

Proof continued.

D	1	3	5	9	15	27	45	135
E	1	39	53	80	69	144	244	423
M	0	40	54	80	68	144	244	423
$E + M$	1	79	107	160	137	288	488	846

(5)

This finite multiplication is itself an executable certificate: start with coefficient one at the group identity; for each factor in (4), replace the coefficient array by the sum of its displayed translates. No generated-group support or unlimited multiplicity substitutes for the exponents.

Put $S_j = \sum_{c,u} \gcd(R, g_c)^j$ and $A_i = 3S_{i+1} - RS_i$. The first five moments from (5) are

$$(S_0, S_1, S_2, S_3, S_4) = (2106, 13962, 315738, 12455226, 722183418).$$

They yield

$$(A_0, A_1, A_2) = (-242424, -937656, -5258952), \quad A_0 A_2 - A_1^2 = 395697405312 > 0.$$

The signed form on linear polynomial coefficients is a positive scalar times a diagonal congruence of $\begin{pmatrix} A_0 & A_1 \\ A_1 & A_2 \end{pmatrix}$. It is negative definite. Substituting $P = 1 + b(x - 1)$ and maximizing the resulting quadratic in b gives (3). Table (5) proves $H = 1$. $\square$

Positive repair one: keep the channel. For a genuine disjoint partition of original states $\mathcal{A} = \bigsqcup A_i$,

$$\boxed{\sum_i \max\{0, \lceil L_d(\mu_i) \rceil\} \leq H.} \quad (6)$$

It strengthens under refinement. Indeed each summand bounds its own integer hit count, and every coarse normalized polynomial is available on each finer piece. Therefore $L_{\text{coarse}} \leq \sum L_{\text{fine}}$; the ceiling inequality and discarding negative summands prove monotonicity.

On the exterior channel of (2), the fixed polynomial $P = (36x - 1)/35$ gives

$$\boxed{\frac{3}{2} \sum_E (x - 1/3) P(x)^2 = \frac{64376}{196875} > 0.} \quad (7)$$

The full coefficient form is $\text{diag}(J_E, J_M)$. The aggregate form is its restriction along $P \mapsto (P, P)$, with pullback $J_E + J_M$. The restriction is not an isomorphism; (7) remains a direction of the original full source even though that diagonal restriction is negative definite. These are disjoint original labels. After other observation maps, the actual mixed Gram must still be transported rather than presumed block diagonal.

Positive repair two: use the discrete spectrum. Define

$$\Phi_j(x) = x \prod_{\nu=1}^j \frac{(2\nu + 1)x - 1}{2\nu}.$$

At $D = 2k + 1 > 1$, this is zero if $k \leq j$ and otherwise equals $(-1)^j \binom{k-1}{j} / D^{j+1}$. Also $\Phi_j(1) = 1$. Thus odd j bounds the original hit indicator below and even j bounds it above. From (5), exact rational evaluation gives

$$\boxed{\sum_{E+M} \Phi_3(1/D) = \frac{1297691}{2460375} > 0, \quad \sum_{E+M} \Phi_4(1/D) = \frac{115374406}{110716875}.} \quad (8)$$

The bracket contains exactly the integer one. Since $48\Phi_3 = 105x^4 - 71x^3 + 15x^2 - x$, positivity follows from the explicit integer inequality $105S_4 - 71RS_3 + 15R^2S_2 - R^3S_1 > 0$. The companion note proves factorial uniform errors for the entire filter family.

The actual witness and its inverse. The sole hit is

$$(c, u, h, r, s, \kappa) = (E, 141879296, 26, 2336, 81289, 341725215823),$$

with original centered exponent vector $(5, -3, -1, 1)$ and raw denominators

$$(4937168704, 722241027794932022, 409884191461517484986392768). \quad (9)$$

Directly $a = hrs$, $u = hr^2$, and $135\kappa = pr + s$. Clearing the reciprocal sum of $(hrs, hsk, phrk)$ gives $(p + 135)\kappa = 4hrs\kappa$, proving ES. Its inverse is $u = a^2/(135Y - pa)$, followed by $(h, r, s) = (d_0^2/u, u/d_0, a/d_0)$ with $d_0 = \gcd(a, u)$. The slope $2336/81289 < 1/8$ lies in the programme's lower positive chamber. No denominator permutation or exponent occurrence has been discarded.

Theorem 2 (All-integer progression with a canonical marked return). *For every integer $j \geq 0$, set*

$$t = 73 + 945j, \quad C = 67632448, \quad n = 4Ct - 135 = 19748674681 + 255650653440j,$$

$$h_0 = 26, \quad r_0 = 32t, \quad s_0 = 81289, \quad a = Ct,$$

$$\kappa_0 = 341725215823 + 8847406287328j + 57265746370560j^2.$$

Then the positive integers $(a, h_0s_0\kappa_0, nh_0r_0\kappa_0)$ solve ES at n . Every n is $1 \pmod{840}$, and infinitely many members are prime.

Proof. Polynomial expansion proves $a = h_0r_0s_0$ and $135\kappa_0 = nr_0 + s_0$. The same reciprocal calculation as for (9) proves the identity; all terms are positive for $j \geq 0$. The initial term and step are coprime, so Dirichlet's theorem supplies infinitely many primes [3, Theorem 18.1]. No hypothesis that t is prime is imposed. If $d = \gcd(r_0, s_0)$, then $\gcd(d, 135) = 1$ because $\gcd(s_0, 135) = 1$. The quotient equation gives $d \mid \kappa_0$. Therefore

$$(h, r, s, \kappa) = (h_0d^2, r_0/d, s_0/d, \kappa_0/d), \quad u = h_0r_0^2$$

is the canonical coprime marking and keeps all three ordered denominators. On this image the parameter inverse is $j = (n - 19748674681)/255650653440$. $\square$

The progression is an explicit specialization of the inherited selector. Its existence is not claimed as a new general progression method, and its later slopes are not all asserted to lie in the separate chamber.

Limits and reproducibility. The counterexample is to a possible universal *converse* for the sufficient first-degree moment test, not to any previously proved sufficient theorem. It is not claimed least. Channel resolution itself is not asserted universally sufficient: the full workbench gives another certified hard-prime shell at which both degree-one channel maxima are negative and the same quartic is positive. No fixed-degree universal sufficiency or ES proof is claimed. The standalone standard-library verifier performs both complete histogram calculations, checks the prime certificates and every labelled inverse, and returns exact rational scores. No probabilistic primality assumption enters.

References

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *J. Aust. Math. Soc.*, 94(1), 50–105, §2.
- [2] The Clankers (2026). *Prime-power boundary kernels and canonical ES count certificates*. ES workbench draft PR6, commit f234ed0c34100c39e52c6a4645793823e830a983, equation M9 and finite screen.
- [3] Sutherland, A. V. (2017). *18.785 Lecture 18: Dirichlet L-functions, primes in arithmetic progressions*, Theorem 18.1.